

SolarCRM

Unified Operations & Lead Lifecycle Management System

Santori Solar Solutions

SYSTEM USER MANUAL

A comprehensive operational guide detailing user roles, access control permissions, lead stages, punching orders, analytics, and recovery operations.

DOCUMENT CONTROLS

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Document Purpose

This user manual serves as the definitive reference document for the SolarCRM system. It is designed to assist administrators, directors,

sales leaders, finance managers, operations teams, and consultants in navigating the web platform and understanding its underlying business logic, authorization rules, and data structures.

By standardizing workflows—from initial lead intake, duplication prevention, and status transition, to order processing

g, document checklist upload s, finance audits, and ops scheduling—this system ensures complete data integrity and coordinates team tasks efficiently.

1. System Introduction & Core Concepts

SolarCRM is a tailored, web-based Customer Relationship Management (CRM) application built specifically to facilitate Santori Solar Solutions' operations. It orchestrates the entire solar client lifecycle, connecting marketing, sales agents, field technical surveyors, finance auditors, and installation supervisors into a unified workspace.

The core architectural philosophy is based on complete accountability. Every action taken on a lead (remarks logged, calls made, status changes, supervisor reassignments) is automatically recorded in audit logs and triggers real-time desktop notifications for affected personnel.

2. User Roles, Designations & Hierarchy

The system defines nine default roles, each mapping to baseline authorization scopes. However, designations are kept separate from functional permissions to support custom titles and unique override sets:

- **Admin & Director:** Full management access. Can create custom user roles, override any permission checklists, view company-wide analytics, and delete records.
- **Sales Head:** Supervises all sales activities. Accesses all leads, schedules meetings, and tracks organizational pipeline performance.
- **Manager:** Manages a specific team branch. Views leads and logs belonging to the Team Leaders and consultants reporting directly to them.
- **Team Leader (TL):** Supervises a pool of Sales/PSA Consultants. Assigns leads and monitors individual caller performance.
- **Finance Manager:** Responsible for verifying order transaction payments, subsidy details, and approving punched sheets.
- **Operations Manager:** Controls scheduling and assigns installers for verified orders, uploading final photos upon completion.
- **Consultant / PSA:** Front-line callers. Allocate leads, book meetings, and punch customer order sheets.

Designation Hierarchy Mapping

Designations dictate supervisor reports. Consultants report to TLs, TLs report to Managers, and Managers report to the Sales Head / Directors. This structure governs automatic lead reassignment: when a consultant is deactivated, their active leads immediately migrate up the hierarchy to their TL to prevent operational gaps.

3. Fine-Grained Permissions Override System

Unlike legacy CRMs that lock access strictly to a user's role/designation, SolarCRM features a user-specific permissions override checklist. Located in the Team Directory edit modal, this option allows administrators to dynamically check or uncheck individual feature permissions for any employee.

Available System Permission Keys

- **leads:view:** Grants visibility to the Leads Pipeline page and sidebar navigation links.
- **leads:view_all:** Allows the user to bypass reporting hierarchy and view all leads in the system.
- **leads:create:** Permits adding new leads manually or importing batch leads from CSV files.
- **leads:edit:** Permits editing core customer details, assigning supervisors, and deactivating leads.
- **leads:change_status:** Enables moving leads through pipeline stages in the details tab.
- **orders:view:** Grants access to the Orders Queue list and document download portals.
- **orders:view_all:** Bypasses hierarchy scoping for order details, allowing company-wide audits.
- **orders:create:** Permits punching order details and uploading Aadhaar/PAN verify documents.
- **orders:verify:** Enables Finance verification controls and Operations scheduler options.
- **orders:submit_installation:** Enables completing orders and uploading installation photos.
- **reports:view:** Enables viewing KPI charts, trends, and consultant performance standing tables.
- **team:view:** Permits accessing the Santori Team directory and viewing member profiles.
- **team:manage:** Enables adding team members, editing roles, and custom permissions overrides.

Backward Compatibility & Absolute-Zero ("none") Access

To preserve backward compatibility, when a user's custom permissions list is left empty (indicated by an empty string "" in the database), the system automatically defaults to the baseline permission set for their designation.

To allow admins to strip all permissions from a user, deselecting all checkboxes explicitly saves the keyword "none" to the database. The system detects "none", blocks default role fallbacks, and restricts the user to viewing only the basic Dashboard shell and their own profile, rendering all other tabs and API endpoints completely inaccessible.

Security Override Principle

Permissions override designations. If a Sales Consultant is checked for "team:manage", they will gain team management panels. If a Team Leader is unchecked for "leads:change_status", they can no longer transition leads, overriding their baseline designation rights.

4. Lead Lifecycle Management & Pipeline Stages

Leads represent prospective customers. They are created manually or imported in bulk from CSV sheets. To protect data quality, the system features a 10-digit mobile duplicate prevention check. If a duplicate mobile number is detected on entry, the system blocks registration and displays the existing Lead ID and its assignee. Administrators or users with edit privileges can choose to bypass this warning and allow the duplicate using the "Override Duplicate" checkbox.

The 13 Sales Pipeline Status Stages

Leads flow through the following stages, regulated by role-based transitions:

- Stage 1: Fresh Lead** - Newly registered lead in the pool, awaiting caller action.
- Stage 2: DNP (No Answer)** - Call attempted but no response. Kept active for follow-up.
- Stage 3: Follow Up** - Client requested a call back. Requires date/time and priority notes.
- Stage 4: Not Interested** - Client declined services. Lead is moved to inactive archive.
- Stage 5: Call Later** - Temporary snooze stage for busy clients.
- Stage 6: Already Installed** - Terminal stage. Client already has solar. Inactive.
- Stage 7: Decision Pending** - Proposal sent; client is reviewing terms and financials.
- Stage 8: Meeting Booked** - Site survey meeting scheduled. Requires meeting date and executive.
- Stage 9: Meeting Done** - Site measurements and structure analysis completed by surveyor.
- Stage 10: Disconnected** - Invalid number or constant network failure.
- Stage 11: Switch Off** - Phone switched off or unreachable.
- Stage 12: Can't Fit Solar** - Terminal stage. Roof structure is unsuitable or shadow-blocked. Inactive.
- Stage 13: Sale Done** - Contract signed and advance payment received. Unlocks the Order tab.

Terminal Stages

Stages 4 (Not Interested), 6 (Already Installed), 12 (Can't Fit Solar), and 13 (Sale Done) mark the end of the active calling lifecycle. Leads in these stages are excluded from active follow-up counters.

5. Order Punching Details & Document Checklist

Once a lead moves to Stage 13 (Sale Done), the "Order Punching & Documents" tab unlocks in their profile page. This tab is used to capture final contractual parameters and upload required verification documents.

Punched Order Details

The consultant must fill out the order forms, specifying details including:

- **System Size (kW):** The capacity of the solar plant in kilowatts.
- **Total Contract Value:** The gross price agreed upon in the sales contract.
- **Payment Terms:** Down payment amount and transaction reference number.
- **Remaining Method:** How the balance will be cleared: Finance/Loan, Cash instalments, or EMI.
- **Subsidy Applicable:** Whether a government subsidy is being claimed, and the calculated amount.

Required Document Uploads Checklist

To process the order, the caller must upload four mandatory files. The system accepts PDF, JPG, and PNG formats (Max size 5MB):

- **Aadhaar Card:** Identity and address proof of the property owner.
- **PAN Card:** Income tax registration verification.
- **Electricity Bill:** Property electricity service provider connection details.
- **Bank Passbook:** First-page copy for subsidy account verification.

Verification & Installation Workflow

Once all documents are uploaded, the order status changes to "Submitted".

- **Finance Audit:** A user with "orders:verify" permission reviews the files and down-payment. Upon approval, they change the order status to "Finance Verified".
 - **Operations Schedule:** Operations schedules the site installation, moving the status to "Ops Assigned".
 - **Completion & Photos:** Upon installation completion, the installer uploads up to 7 completion photos. The status is updated to "Completed", locking the order records.
-

6. KPI Analytics & Performance Dashboard

SolarCRM includes an analytics module that transforms database logs into actionable charts and leaderboard tables. Access to reports is governed by the "reports:view" permission override.

Dashboard KPI Cards

The main dashboard displays operational metrics, including:

- **Total Leads Pool:** Total number of active and inactive leads within the user's access scope.
- **Active Leads:** Leads currently undergoing active calling (excluding terminal statuses).
- **Meetings Booked:** Total site surveys scheduled for the current calendar month.
- **Sales Closed:** Number of signed deals (Stage 13) recorded this month.
- **Conversion Rate:** The percentage of allocated leads successfully closed as sales.

Trend Analysis Charts

Interactive charts plot lead generation vs closure rates over the last 15 days, helping managers identify sales bottlenecks.

Consultant Performance Leaderboards

Consultant Standings Table Headers	
Column Header	Description / Metric Tracked
Leads Allocated	Total leads assigned to the consultant in their history.
Nurture Actions	Number of follow-up remarks and status updates logged.
Site Meetings	Total site survey meetings scheduled or completed.
Sales Closed	Total leads successfully moved to Stage 13 (Sale Done).

CSV Spreadsheet Exporter

Users with export permissions can download the entire lead database as a CSV spreadsheet directly from the interface, facilitating external audits and marketing operations.

7. Security, Git Isolation & Admin Recovery

System security is managed through database configuration, file-system isolation, and terminal utilities.

Git Uploads Isolation

Because users upload sensitive customer documentation (PAN/Aadhaar) and call audio recordings, the entire `/uploads` folder is added to `.gitignore`. This prevents files from being pushed to Git repositories, ensuring private customer records are kept secure on local server storage.

Admin User Password Resets

Admins can reset any user's password directly from the team edit modal. The new password is encrypted using `bcryptjs` (salt rounds: 10) before being saved to the database. This reset action requires `team:manage` permissions.

Emergency Password Recovery (Admin Forgot Password)

If the system administrator forgets their password, they cannot use the UI (since password reset is gated inside the authenticated dashboard). To resolve this, a secure terminal utility script is provided:

```
// Reset any user password directly from the server hosting terminal  
  
npx tsx scripts/reset-password.ts <admin_email> <new_password>
```

How the Recovery Script Works

- **Prisma Integration:** The script instantiates a direct Prisma client to connect to the PostgreSQL instance.
- **Credential Hashing:** It hashes the new password with bcrypt (salt rounds: 10) to match CRM encryption.
- **User Update:** It updates the target user's password hash in the database and terminates the session, forcing re-authentication on the next login attempt.

Terminal Execution Safety

The recovery script can only be run by users who have SSH/terminal access to the hosting server, keeping database operations secure.

